United States Patent Application Publication 20250280692
Kind Code A1
Publication Date September 04, 2025
Inventor(s) YANG; Zhi et al.

PIXEL STRUCTURE AND DISPLAY PANEL

Abstract

A pixel structure and a display panel are provided. The pixel structure includes a first sub-pixel, a second sub-pixel and a third sub-pixel. The first sub-pixel and the second sub-pixel are alternately disposed along a first direction and form a first pixel row. The third sub-pixels are disposed along the first direction and form a second pixel row. The first pixel row and the second pixel row are alternately disposed along a second direction. Centers of two first sub-pixels and two second sub-pixels from adjacent two rows and adjacent two columns may be connected to form a virtual quadrilateral. The third sub-pixel is located within the virtual quadrilateral. The pixel structure further includes spacer structures. Each spacer structure is configured to enclose one sub-pixel. The spacer structure is configured to electrically connect the cathodes of the sub-pixels to each other.

Inventors: **YANG; Zhi (SHENZHEN, CN), TANG; Yangling (SHENZHEN, CN), XIE; Zhisheng (SHENZHEN, CN), JIANG; Chenchen (SHENZHEN, CN), LIANG; Qin (SHENZHEN, CN), ZHOU; Xiufeng (SHENZHEN, CN), YUAN; Haijiang (SHENZHEN, CN)**

Applicant: **HKC CORPORATION LIMITED** (SHENZHEN, CN)

Family ID: **91407784**

Assignee: **HKC CORPORATION LIMITED (SHENZHEN, CN)**

Appl. No.: **19/030403**

Filed: **January 17, 2025**

Foreign Application Priority Data

CN	202410235506.4	Feb. 29, 2024

Publication Classification

Int. Cl.: **H10K59/35** (20230101); **H10K59/122** (20230101); **H10K59/80** (20230101)

U.S. Cl.:

CPC **H10K59/353** (20230201); **H10K59/122** (20230201); **H10K59/352** (20230201); **H10K59/8052** (20230201);

Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS

[0001] This application claims priority to Chinese Patent Application No. 202410235506.4, filed Feb. 29, 2024, which is herein incorporated by reference in its entirety.

TECHNICAL FIELD

[0002] The present disclosure relates to the field of display technologies, and in particular to a pixel structure and a display panel.

BACKGROUND

[0003] Nowadays, in conventional manufacturing methods of organic light-emitting diodes (OLEDs), the material vaporization regions are defined by fine metal mask (FMM). However, due to issues such as common color mixing, product designers must ensure generally too large PDL gap values or pixel pitches, which may result in reduction of product apertures. The product lifespan capacity may be directly affected.

SUMMARY OF THE DISCLOSURE

[0004] A first technical scheme adopted by the present disclosure is to provide a pixel structure. The pixel structure includes a first sub-pixel, a second sub-pixel and a third sub-pixel. The colors of the first sub-pixel, the second sub-pixel and the third sub-pixel are different from each other. The first sub-pixel and the second sub-pixel are alternately disposed along a first direction and form a first pixel row. A plurality of the third sub-pixels are disposed along the first direction and form a second pixel row. The first pixel row and the second pixel row are alternately disposed along a second direction, and the first direction and the second direction are perpendicular to each other. Two first sub-pixels and two second sub-pixels from adjacent two rows and adjacent two columns are arranged in a 2×2 matrix, the two first sub-pixels are located in different rows and different columns, the two second sub-pixels are located in different rows and different columns. Two centers of the two first sub-pixels and the two centers of the two second sub-pixels are capable of being connected to form a virtual quadrilateral. The third sub-pixel is located within the virtual quadrilateral. A side of the third sub-pixel is parallel to a proximate side of an adjacent sub-pixels. The pixel structure further includes spacer structures. The spacer structures are configured to enclose the sub-pixels, and are in one-to-one correspondence with the sub-pixels. The spacer structure is configured to electrically connect the cathodes of the sub-pixels to each other.

[0005] In some embodiments, each of the first sub-pixel and the second sub-pixel is a rhombus, the third sub-pixel is a parallelogram. A diagonal line of the rhombus is configured to extend in the first direction, another diagonal line of the rhombus is configured to extend in the second direction. The first sub-pixel and the second sub-pixel adjacent to each other along the first direction are configured to be disposed as corner-facing-corner, and the first sub-pixel and the second sub-pixel adjacent to each other along the second direction are configured to be disposed as corner-facing-corner. A side of the spacer structure is parallel to a proximate side of a sub-pixel adjacent to the spacer structure. The first sub-pixel and the third sub-pixel adjacent to the first sub-pixel share a side wall of the spacer structure, and the second sub-pixel and the third sub-pixel adjacent to the second sub-pixel share a side wall of the spacer structure.

[0006] In some embodiments, the region enclosed by the spacer structure is polygonal. The corners of the region enclosed by the spacer structure configured to enclose the first sub-pixel are flat

chamfers or rounded chamfers, and the corners of the region enclosed by the spacer structure configured to enclose the second sub-pixel are flat chamfers or rounded chamfers.
[0007] In some embodiments, the spacer structure includes a spacer part and an eave structure. The eave structure is provided on an upper surface of the spacer part. The eave structure is configured to cover the spacer part, and extend beyond the spacer part in a direction parallel to the spacer part. The cross-section of the spacer structure along a plane parallel to the eave structure is configured to taper along a direction toward the eave structure.
[0008] In some embodiments, for each spacer structure, at least a part of the side wall is a first side wall. The spacer part of the first side wall is electrically conductive, and is configured to contact the cathode of the sub-pixel adjacent to the first side wall, and achieve an electrical connection between the spacer structure and the sub-pixel. The cathode of the sub-pixel is vaporized onto a surface of the light-emitting layer of the sub-pixel by a vaporization source. The first direction is a long-edge direction of the vaporization source, and the second direction is a scanning direction of the vaporization source.
[0009] In some embodiments, a part of the side walls of each spacer structure is the first side wall, and another part of the side walls of the each spacer structure is a second side wall. The spacer part of the second side wall is insulating.
[0010] In some embodiments, the spacer structure configured to enclose the first sub-pixel is referred to as a first spacer structure, and the spacer structure configured to enclose the second sub-pixel is referred to as a second spacer structure. A straight line on which the diagonal line of the rhombus extending in the first direction is referred to as a first symmetry line, and a straight line on which the diagonal line of the rhombus extending in the second direction is referred to as a second symmetry line. For the first spacer structure and the second spacer structure adjacent to each other along the first direction, the first side wall of the first spacer structure and the first side wall of the second spacer structure are disposed at opposite sides of the corresponding first symmetry line, the first side walls adjacent to each other along the first direction are configured to contact each other and are electrically connected. For the first spacer structure and the second spacer structure adjacent to each other along the second direction: the first side wall of the first spacer structure and the first side wall of the second spacer structure are disposed at the same side of the respectively corresponding first symmetry lines; or the first side wall of the first spacer structure and the first side wall of the second spacer structure are disposed at opposite sides of the respectively corresponding first symmetry lines.
[0011] In some embodiments, the spacer structure configured to enclose the first sub-pixel is referred to as a first spacer structure, and the spacer structure configured to enclose the second sub-pixel is referred to as a second spacer structure. A straight line on which the diagonal line of the rhombus extending in the first direction is referred to as a first symmetry line, and a straight line on which the diagonal line of the rhombus extending in the second direction is referred to as a second symmetry line. For the first spacer structure and the second spacer structure adjacent to each other along the second direction, the first side wall of the first spacer structure and the first side wall of the second spacer structure are disposed at opposite sides of the corresponding second symmetry line; the first side walls adjacent to each other along the second direction are configured to contact each other and are electrically connected. For the first spacer structure and the second spacer structure adjacent to each other along the first direction: the first side wall of the first spacer structure and the first side wall of the second spacer structure are disposed at the same side of the respectively corresponding second symmetry lines; or the first side wall of the first spacer structure and the first side wall of the second spacer structure are disposed at opposite sides of the respectively corresponding second symmetry lines.
[0012] In some embodiments, the side walls of a part of the first spacer structures are all first side walls; and/or, the side walls of a part of the second spacer structures are all first side walls.
[0013] A second technical solution adopted by the present disclosure is to provide a display panel.

The display panel includes the pixel structure. The pixel structure includes a first sub-pixel, a second sub-pixel and a third sub-pixel. The colors of the first sub-pixel, the second sub-pixel and the third sub-pixel are different from each other. The first sub-pixel and the second sub-pixel are alternately disposed along a first direction and form a first pixel row. A plurality of the third sub-pixels are disposed along the first direction and form a second pixel row. The first pixel row and the second pixel row are alternately disposed along a second direction, and the first direction and the second direction are perpendicular to each other. Two first sub-pixels and two second sub-pixels from adjacent two rows and adjacent two columns are arranged in a 2×2 matrix, the two first sub-pixels are located in different rows and different columns, the two second sub-pixels are located in different rows and different columns. Two centers of the two first sub-pixels and the two centers of the two second sub-pixels are capable of being connected to form a virtual quadrilateral. The third sub-pixel is located within the virtual quadrilateral. A side of the third sub-pixel is parallel to a proximate side of an adjacent sub-pixels. The pixel structure further includes spacer structures. The spacer structures are configured to enclose the sub-pixels, and are in one-to-one correspondence with the sub-pixels. The spacer structure is configured to electrically connect the cathodes of the sub-pixels to each other.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0014] To more clearly illustrate technical solutions in the present disclosure, the drawings required in the description of the embodiments will be briefly introduced below. Obviously, the drawings in the following description are only some embodiments of the present disclosure. For those of ordinary skills in the art, other drawings could be obtained based on these drawings without creative efforts.

[0015] FIG. **1** is a schematic structural diagram of a pixel structure according to a first embodiment of the present disclosure.

[0016] FIG. **2** is a schematic sectional structural diagram along a direction E-E of FIG. **1**.

[0017] FIG. **3** is a schematic structural diagram of a pixel structure according to a second embodiment of the present disclosure.

[0018] FIG. **4** is a schematic structural diagram of a pixel structure according to a third embodiment of the present disclosure.

[0019] FIG. **5** is a schematic structural diagram of a pixel structure according to a fourth embodiment of the present disclosure.

[0020] FIG. **6** is a schematic structural diagram of a pixel structure according to a fifth embodiment of the present disclosure.

[0021] FIG. **7** is a schematic structural diagram of a pixel structure according to a sixth embodiment of the present disclosure.

[0022] FIG. **8** is a schematic structural diagram of a pixel structure according to a seventh embodiment of the present disclosure.

[0023] FIG. **9** is a schematic structural diagram of a pixel structure according to an eighth embodiment of the present disclosure.

[0024] FIG. **10** is a schematic structural diagram of a pixel structure according to a ninth embodiment of the present disclosure.

[0025] FIG. **11** is a schematic structural diagram of a pixel structure according to a tenth embodiment of the present disclosure.

[0026] FIG. **12** is a schematic structural diagram of a pixel structure according to an eleventh embodiment of the present disclosure.

[0027] FIG. **13** is a schematic structural diagram of a pixel structure according to a twelfth

embodiment of the present disclosure.
[0028] FIG. **14** is a schematic structural diagram of a pixel structure according to a thirteenth embodiment of the present disclosure.
[0029] FIG. **15** is a schematic comparison diagram of a sub-pixel life decay process according to an embodiment of the present disclosure.

REFERENCE NUMBER

[0030] **10**, sub-pixel; **11**, anode; **12**, light-emitting layer; **13**, cathode; **10**A, first sub-pixel; **10**B, second sub-pixel; **10**C, third sub-pixel; **110**, first pixel row; **120**, second pixel row; X, first direction; Y, second direction; D**1**, first symmetry line; D**2**, second symmetry line; G**1**, first side; G**2**, second side; G**3**, third side; G**4**, fourth side; **20**, spacer structure; **21**, side wall; **21**A, first side wall; **21**B, second side wall; **211**, spacer part; **212**, eave structure; **30**, pixel definition layer; **31**, pixel aperture.

DETAILED DESCRIPTION

[0031] Technical solutions of embodiments of the present disclosure are described in detail below in conjunction with the accompanying drawings.
[0032] In the following description, specific details such as particular system structures, interfaces, techniques, etc. are presented for the purpose of illustration and not for the purpose of limitation, to facilitate a thorough understanding of the present disclosure.
[0033] The technical solutions in embodiments of the present disclosure will be described clearly and thoroughly in connection with accompanying drawings of the embodiments of the present disclosure. Obviously, the described embodiments are only a part of the embodiments, but not all of them. All other embodiments obtained by a person of ordinary skills in the art based on embodiments of the present disclosure without creative efforts should all be within the protection scope of the present disclosure.
[0034] The terms ‘first’, ‘second’, and ‘third’ in this disclosure are only for the purpose of description, and cannot be construed as indicating or implying relative importance or implicitly indicating the number of technical features referred to. Therefore, the features defined with ‘first’, ‘second’, and ‘third’ may explicitly or implicitly include at least one of the features. In the description of the present disclosure, ‘a plurality of’ means at least two, such as two, three, etc., unless otherwise specifically defined. All directional indicators (such as up, down, left, right, front, back . . .) in embodiments of the present disclosure are only used to explain a motion state, a relative positional relationship between the components in a specific posture (as illustrated in the drawings). If the specific posture changes, then the directional indication will change accordingly. In addition, the terms ‘include’, ‘comprise’ and any variations thereof are intended to cover non-exclusive inclusion. For example, a process, a method, a system, a product, or a device that includes a series of operations or units is not limited to the listed operations or units, but optionally includes unlisted operations or units, or optionally also includes other operations or units inherent to these processes, methods, products or devices.
[0035] Reference to ‘embodiment’ herein means that, a specific feature, structure, or characteristic described in conjunction with the embodiments may be included in at least one embodiment of the present disclosure. The appearance of this phrase in various locations of the specification does not necessarily refer to the same embodiment, nor is it an independent or alternative embodiment mutually exclusive with other embodiments. Those skilled in the art may explicitly and implicitly understand that, the embodiments described herein may be combined with other embodiments.
[0036] As illustrated in FIG. **1** and FIG. **2**, FIG. **1** is a schematic structural diagram of a pixel structure according a first embodiment of the present disclosure, and FIG. **2** is a schematic sectional structural diagram along a direction E-E of FIG. **1**.
[0037] A pixel structure is provided in the present disclosure. The pixel structure includes a first sub-pixel **10**A, a second sub-pixel **10**B and a third sub-pixel **10**C. The colors of the first sub-pixel **10**A, the second sub-pixel **10**B and the third sub-pixel **10**C are different from each other. The first

sub-pixel **10**A and the second sub-pixel **10**B are alternately disposed along a first direction X and form a first pixel row **110**. A plurality of third sub-pixels **10**C are disposed along the first direction X and form the second pixel row **120**. The first pixel row **110** and the second pixel row **120** are alternately disposed along a second direction Y. The first direction X is perpendicular to the second direction Y. Two first sub-pixels **10**A and two second sub-pixels **10**B from two adjacent rows and two adjacent columns are arranged in a 2×2 matrix. The two first sub-pixels **10**A are located in different rows and in different columns, and the two second sub-pixels **10**B are located in different rows and in different columns. Two centers of the two first sub-pixels **10**A and two centers of the two second sub-pixels **10**B are capable of being connected as four vertices to form a virtual quadrilateral. The third sub-pixel **10**C is located within this virtual quadrilateral. A side of the third sub-pixel **10**C is parallel to a proximate side of a sub-pixel **10** adjacent to the third sub-pixel **10**C. The pixel structure further includes a spacer structure **20**. The spacer structures **20** are each configured to enclose the sub-pixel **10**, and are in one-to-one correspondence with the sub-pixels **10**. The spacer structure **20** is configured to electrically connect cathodes **13** of the sub-pixels **10** to each other.
[0038] In the present disclosure, the spacer structure **20** is configured to enclose the sub-pixel **10**. In this way, the spacer structure **20** may isolate different sub-pixels **10** and prevent pixel crosstalk. The spacer structure **20** may also enable mutual electrical connection between the cathodes **13** of the sub-pixels **10**, thereby facilitating the electrical interconnection of the cathodes **13** of the sub-pixels **10** of a whole panel. In addition, a fine metal mask is adopted in the related technology to define a material vaporization region. In contrast, the spacer structure **20** in the present disclosure may further reduce a spacing between the sub-pixels **10**, effectively increase an area of the pixel aperture **31**, and reduce current density of the pixels. In this way, a product lifespan may be effectively increased, and production cost of the pixel structure may also be reduced.
[0039] A pixel definition layer **30** is also provided in the present disclosure. The pixel definition layer **30** defines a location for the sub-pixel **10**. Specifically, a plurality of pixel apertures **31** are defined in the pixel definition layer **30**. The sub-pixel **10** is disposed within the pixel aperture **31**. The pixel apertures **31** are disposed in one-to-one correspondence with the sub-pixels **10**.
[0040] The material and thickness of the pixel definition layer **30** are not limited herein, and may be selected per actual requirements.
[0041] In the present disclosure, the first sub-pixel **10**A, the second sub-pixel **10**B and the third sub-pixel **10**C represents sub-pixels **10** of different colors.
[0042] The sub-pixel **10** includes an anode **11**, a light-emitting layer **12**, and a cathode **13** cascading disposed or stacked in sequence. The materials of the anode **11**, the light-emitting layer **12**, and the cathode **13** are not limited herein, and may be selected per actual requirements. The cathode **13** of the sub-pixel **10** is vaporized onto a surface of the light-emitting layer **12** of the sub-pixel **10** by a vaporization source. The first direction X is a long-edge direction of the vaporization source, and the second direction Y is a scanning direction of the vaporization source.
[0043] The sub-pixel **10** includes an OLED.
[0044] In some embodiments, the first sub-pixel **10**A is a red sub-pixel **10**. The second sub-pixel **10**B is a blue sub-pixel **10**. The third sub-pixel **10**C is a green sub-pixel **10**.
[0045] In some other embodiments, the first sub-pixel **10**A, the second sub-pixel **10**B, and the third sub-pixel **10**C may respectively represent sub-pixels **10** of other colors. The colors of the sub-pixels are not limited herein and may be selected per actual requirements. In the present disclosure, the **10** relationships among sizes of the light-emitting areas of the sub-pixels **10** of various colors are not limited herein, and may be selected per actual requirements.
[0046] A spacing between the third sub-pixel **10**C and the second sub-pixel **10**B may be greater than, equal to, or less than a spacing between the third sub-pixel **10**C and the first sub-pixel **10**A, which may be determined per actual requirements.
[0047] In some embodiments, the spacing between the third sub-pixel **10**C and the second sub-

pixel **10**B is equal to the spacing between the third sub-pixel **10**C and the first sub-pixel **10**A, which would guarantee a display homogeneity.

[0048] The first direction X is a row direction of the sub-pixel **10**. The second direction Y is a column direction of the sub-pixel **10**.

[0049] For two first pixel rows **110** adjacent to each other along the second direction Y, an alternating order in the first direction X between the first sub-pixel **10**A and the second sub-pixel **10**B of one of the two first pixel rows **110** is opposite to an alternating order in the first direction X between the first sub-pixel **10**A and the second sub-pixel **10**B of another one of the two first pixel row **110**. In some embodiments, for two adjacent first pixel rows **110**: in one first pixel row **110**, the first sub-pixel **10**A and the second sub-pixel **10**B are alternately arranged in sequence along the first direction X; and in another first pixel row **110**, the second sub-pixel **10**B and the first sub-pixel **10**A are alternately arranged in sequence along the first direction X.

[0050] A side of the third sub-pixel **10**C is parallel to a proximate side of the sub-pixel **10** adjacent to the third sub-pixel **10**C. In other words, a side edge of the third sub-pixel **10**C proximate to an adjacent sub-pixel **10** is parallel to a side edge of this adjacent sub-pixel **10** proximate to the third sub-pixel **10**C. Specifically, the side of the third sub-pixel **10**C and a proximate side of the adjacent first sub-pixel **10**A are in parallel with each other, the side of the third sub-pixel **10**C and a proximate side of the adjacent second sub-pixel **10**B are in parallel with each other.

[0051] In some embodiments, each of the first sub-pixel **10**A and the second sub-pixel **10**B is a rhombus. The third sub-pixel **10**C is a parallelogram. One diagonal line of the rhombus is configured to extend in the first direction X, another diagonal line of the rhombus is configured to extend in the second direction Y. The first sub-pixel **10**A and the second sub-pixel **10**B adjacent to each other in the first direction X are disposed to be corner-facing-corner. The term “corner-facing-corner” means that a corner of a sub-pixel **10** is facing or against a corner of another sub-pixel **10**. The first sub-pixel **10**A and the second sub-pixel **10**B adjacent to each other in the second direction Y are disposed as corner-facing-corner. In other words, in each first pixel row **110**, a diagonal line of the first sub-pixel **10**A extending along the first direction X and a diagonal line of the second sub-pixel **10**B extending along the first direction X are on a same straight line, such that the first sub-pixel **10**A and the second sub-pixel **10**B are disposed as corner-facing-corner in the first direction X. Similarly, for the first sub-pixel **10**A and the second sub-pixel **10**B that are adjacent to each other in the second direction Y, a diagonal line of the first sub-pixel **10**A extending along the second direction Y and the diagonal line of the second sub-pixel **10**B extending along the second direction Y are one a same straight line, such that the first sub-pixel **10**A and the second sub-pixel **10**B are disposed as corner-facing-corner in the second direction Y.

[0052] In some embodiments, provided that the side of the third sub-pixel **10**C is parallel to the proximate side of the adjacent sub-pixel **10**, the third sub-pixel **10**C may also be other kind of polygons, such as a hexagon.

[0053] In the present disclosure, there is a gap between the first sub-pixel **10**A and the second sub-pixel **10**B adjacent to each other along the first direction X or the second direction Y.

[0054] In some embodiments, the extension direction of each of the side edges of all the sub-pixels **10** is configured to intersect with both the first direction X and the second direction Y. That is, the extension direction of any of the side edges of the sub-pixel **10** is neither parallel to the first direction X nor parallel to the second direction Y.

[0055] The spacer structure **20** is a ring-shaped structure, and is configured to protrude from the pixel definition layer **30**. The spacer structure **20** is configured to enclose the pixel aperture **31**, so as to enclose the sub-pixels **10**. A single spacer structure **20** encloses a single pixel aperture **31**.

[0056] Further, the spacer structure **20** is parallel to a proximate side of an adjacent sub-pixel **10**. In other words, the side edge of the spacer structure **20** proximate to the adjacent sub-pixel **10** and the side edge of the adjacent sub-pixel **10** proximate to the spacer structure **20** are parallel to each other, thereby facilitating a regular preparation of the spacer structure **20**.

[0057] The first sub-pixel **10**A and an adjacent third sub-pixel **10**C share a side wall **21** of the spacer structure **20**. The second sub-pixel **10**B and the adjacent third sub-pixel **10**C share another side wall **21** of the spacer structure **20**. In other words, the spacer structure **20** enclosing the first sub-pixel **10**A shares a same side wall **21** with the spacer structure **20** enclosing the third sub-pixel **10**C adjacent to this first sub-pixel **10**A, thereby reducing the spacing between the sub-pixels **10** and beneficially increasing the pixel aperture ratio.
[0058] The region enclosed by the spacer structure **20** is polygonal. The corners of the region enclosed by the spacer structure **20** configured to enclose the first sub-pixel **10**A are flat chamfers or rounded chamfers. The corners of the region enclosed by the spacer structure **20** configured to enclose the second sub-pixel **10**B are flat chamfers or rounded chamfers. In the present disclosure, by removing sharp corners of the spacer structures **20** for enclosing the first sub-pixel **10**A and the second sub-pixel **10**B, the effect of the irregularity of the sharp corners on the area size of the pixel aperture **31** may be eliminated, thereby facilitating the reduction of the spacing between the first sub-pixel **10**A and the second sub-pixel **10**B, and further increasing the pixel aperture ratio. Furthermore, by this way, the problem of poor attachment between the spacer structure **20** and the sub-pixel **10** may also be avoided. The cathode **13** of the sub-pixel **10** at a sharp corner is prone to a problem of poor attachment with the spacer structure **20**.
[0059] In some embodiments, the corners of the region enclosed by the spacer structure **20** configured to enclose the first sub-pixel **10**A are flat chamfers. The corners of the region enclosed by the spacer structure **20** configured to enclose the second sub-pixel **10**B are flat chamfers.
[0060] In some embodiments, the corners of the region enclosed by the spacer structure **20** configured to enclose the third sub-pixel **10**C are flat chamfers or rounded chamfers, which is not unnecessarily limited herein. The specific arrangement of the corners may be selected per actual requirements.
[0061] The spacer structure **20** includes a spacer part **211** and an eave structure **212**. The eave structure **212** is provided on an upper surface of the spacer part **211**. The eave structure **212** is configured to cover or shield the spacer part **211**, and extends beyond the spacer part **211** in a direction parallel to the spacer part **211**. In other words, an orthographic projection of the eave structure **212** onto the pixel definition layer **30** overlaps or covers the orthographic projection of the spacer part **211** onto the pixel definition layer **30**, and an area of the orthographic projection of the eave structure **212** onto the pixel definition layer **30** is greater than that of the orthographic projection of the spacer part **211** onto the pixel definition layer **30**. During the vaporization process of the sub-pixel **10**, a vaporization angle of the vaporized material may be adjusted by edges of the eave structure **212**, so that the cathode **13** may cover the light-emitting layer **12** and achieve a good attachment between the cathode **13** and the spacer structure **20**.
[0062] The spacer part **211** of the spacer structure **20** is attached to the cathode **13** of the enclosed sub-pixel **10**.
[0063] A cross-section of the spacer structure **20** along a plane parallel to the eave structure **212** is configured to taper along a direction toward the eave structure **212**. In other words, an angle between a side face of the spacer part **211** and a portion of the eave structure **212** extending beyond the spacer part **211** is less than 90 degrees. In this way, an attachment effect of the cathode **13** of the sub-pixel **10** to the side face of the spacer part **211** may be enhanced, and thereby facilitating a sound attachment of the cathode **13** of the sub-pixel **10** to the spacer part **211**.
[0064] For each spacer structure **20**, at least a portion of the side wall **21** is a first side wall **21**A. The spacer part **211** of the first side wall **21**A is electrically conductive, and configured to contact the cathode **13** of an adjacent sub-pixel **10**, so as to achieve an electrical connection between the spacer structure **20** and the sub-pixel **10**. In other words, each spacer structure **20** includes a plurality of side walls **21**. Each side wall **21** corresponds to a side edge of the enclosed sub-pixel **10**. A part of or all the side walls **21** in each spacer structure **20** are the first side wall **21**A. The spacer part **211** of the first side wall **21**A of the spacer structure **20** is attached to the cathode **13** of

the adjacent sub-pixel **10**, so as to achieve the electrical connection between this spacer structure **20** and the corresponding sub-pixel **10**.

[0065] In the present disclosure, the first direction X is configured as the long-edge direction of the vaporization source, and the second direction Y is configured as the scanning direction of the vaporization source. In this way, any first side wall **21**A of the spacer structure **20** is enabled to attach with the cathode **13** of the sub-pixel **10**, thereby achieving the electrical connection between the spacer structure **20** and the adjacent sub-pixel **10**.

[0066] In some embodiments, a part of the side wall **21** of each spacer structure **20** is the first side wall **21**A and another part of the side wall **21** of each spacer structure **20** is a second side wall **21**B. The spacer part **211** of the second side wall **21**B is insulating. The eave structure **212** is insulating.

[0067] The wall thickness value of the first side wall **21**A and the wall thickness value of the second side wall **21**B are not limited herein, and may be selected per actual requirements. The wall thickness of the first side wall **21**A may be greater than or equal to, or less than, the wall thickness of the second side wall **21**B.

[0068] The material of the spacer part **211** of the second side wall **21**B may be the same as or different from that of the eave structure **212**. These materials are not unnecessarily limited herein and may be selected per actual requirements.

[0069] In some embodiments, the spacer structure **20** configured to enclose the first sub-pixel **10**A is defined as a first spacer structure, and the spacer structure **20** configured to enclose the second sub-pixel **10**B is defined as a second spacer structure. A straight line on which the diagonal line of the rhombus in the first direction X extends is defined as a first symmetry line D**1**, and a straight line on which the diagonal line of the rhombus in the second direction Y extends is defined as a second symmetry line D**2**. In other words, there is a first symmetry line D**1** in each first pixel rows **110**, and there is a second symmetry line D**2** in each column including the first sub-pixels **10**A and the second sub-pixels **10**B.

[0070] For a first spacer structure and a second spacer structure adjacent to each other along the first direction X, the first side wall **21**A of this first spacer structure and the first side wall **21**A of this second spacer structure are disposed at opposite sides of the corresponding first symmetry line D**1**. The corresponding first symmetry line D**1** that corresponds to the first spacer structure is the first symmetry line of the first pixel row **110** including the first sub-pixel **10**A enclosed by this first spacer structure. The corresponding first symmetry line D**1** that corresponds to the second spacer structure is the first symmetry line of the first pixel row **110** including the second sub-pixel **10**B enclosed by this second spacer structure. The first side walls **21**A adjacent to each other along the first direction X are configured to contact with each other and are electrically connected. By this arrangement manner of the first side walls **21**A, the cathodes **13** of the sub-pixels **10** from adjacent first pixel row **110** and second pixel row **120** may be mutually electrically connected through the first side walls **21**A, thereby realizing mutual electrical connection between the cathodes **13** of the sub-pixels **10** of the pixel structure, and forming a whole panel connection of the cathodes **13**, which is beneficial to the homogeneity of the cathodes **13**.

[0071] For a first spacer structure and a second spacer structure adjacent to each other along the second direction Y: the first side wall **21**A of this first spacer structure and the first side wall **21**A of the second spacer structure are disposed at a same side of the corresponding first symmetry lines D**1** respectively; or, the first side wall **21**A of this first spacer structure and the first side wall **21**A of this second spacer structure are disposed on different sides of the corresponding first symmetry lines D**1** respectively.

[0072] In some embodiments, the first symmetry line D**1** includes a first side G**1** and a second side G**2** disposed opposite to each other along the second direction Y. The first side G**1** corresponding to each first symmetry line D**1** is disposed at a same side of the second direction Y. For example, as illustrated in FIG. **1**, the first side G**1** is on a side pointed by the arrow of the second direction Y, or vice versa.

[0073] In some embodiments, the first side wall **21**A of the first spacer structure and the first side wall **21**A of the second spacer structure adjacent to the first spacer structure along the second direction Y are both disposed on a first side G**1** of a corresponding first symmetry line D**1**.

[0074] As illustrated in FIG. **1** to FIG. **3**, FIG. **3** is a schematic structural diagram of a pixel structure according a second embodiment of the present disclosure.

[0075] In some embodiments, as illustrated in FIG. **3**, for a first spacer structure and a second spacer structure adjacent to each other along the second direction Y, the first side wall **21**A of this first spacer structure is disposed on the second side G**2** of the first symmetry line D**1** corresponding to the first spacer structure, and the first side wall **21**A of this second spacer structure is also disposed on the second side G**2** of a first symmetry line D**1** corresponding to the second spacer structure.

[0076] As illustrated in FIG. **1**, FIG. **2**, FIG. **4** to FIG. **5**, FIG. **4** is a schematic structural diagram of a pixel structure according a third embodiment of the present disclosure. FIG. **5** is a schematic structural diagram of a pixel structure according a fourth embodiment of the present disclosure.

[0077] In some embodiments, for the first spacer structure and the second spacer structure adjacent to each other along the second direction Y, the first side wall **21**A of the first spacer structure and the first side wall **21**A of the second spacer structure are disposed on different sides of the corresponding first symmetry lines D**1** respectively. In some embodiments, as illustrated in FIG. **4**, the first side wall **21**A of the first spacer structure is disposed on the first side G**1** of the first symmetry line D**1** corresponding to the first spacer structure, and the first side wall **21**A of the second spacer structure is disposed on the second side G**2** of the first symmetry line D**1** corresponding to the second spacer structure. In some embodiments, as illustrated in FIG. **5**, the first side wall **21**A of the first spacer structure is disposed on the second side G**2** of the first symmetry line D**1** corresponding to the first spacer structure, and the first side wall **21**A of the second spacer structure is disposed on the first side G**1** of the first symmetry line D**1** corresponding to the second spacer structure.

[0078] As illustrated in FIG. **1**, FIG. **2** and FIG. **6**, FIG. **6** is a schematic structural diagram of a pixel structure according a fifth embodiment of the present disclosure.

[0079] A structure of the fifth embodiment of the present disclosure is substantially similar to that of the first embodiment. The difference between the fifth embodiment and the first embodiment is that: the side walls **21** of a part of or some the second spacer structures are all first side walls **21**A.

[0080] In some embodiments, the side walls **21** of a part of the first spacer structures are all first side walls **21**A, and/or, the side walls **21** of a part of the second spacer structures are all first side walls **21**A. By providing all the side walls **21** of a part of the first spacer structures as the first side wall **21**A, the attachment area between the first spacer structure and the cathode **13** of the first sub-pixel **10**A enclosed by this first spacer structure may be increased, thereby enhancing the electrical connection effect between this first spacer structure and the cathode **13** of the corresponding first sub-pixel **10**A. The attachment area between the first spacer structure and the cathode **13** of the third sub-pixel **10**C adjacent to the first spacer structure may also be increased, thereby enhancing the electrical connection effect between the first spacer structure and the cathode **13** of the adjacent third sub-pixel **10**C. Similarly, by providing all the side walls **21** of a part of the second spacer structures as the first side wall **21**A, an attachment area between the second spacer structure and the cathode **13** of the second sub-pixel **10**B enclosed by this second spacer structure may be increased, thereby enhancing the electrical connection effect between this second spacer structure and the cathode **13** of the second sub-pixel **10**B. The attachment area between the second spacer structure and the cathode **13** of an third sub-pixel **10**C adjacent to the second spacer structure may also be increased, thereby enhancing the electrical connection effect between the second spacer structure and the cathode **13** of the adjacent third sub-pixel **10**C.

[0081] In the present disclosure, “A and/or B” means only A is included, or only B is included, or both A and B are included.

[0082] In some embodiments, the side walls **21** of a part of the second spacer structures are all first side walls **21**A. In some embodiments, all the side walls **21** of the second spacer structure corresponding to or enclosing the second sub-pixel **10**B in the even-numbered first pixel rows **110** are the first side walls **21**A.
[0083] As illustrated in FIG. **1**, FIG. **2** and FIG. **7**, FIG. **7** is a schematic structural diagram of a pixel structure according a sixth embodiment of the present disclosure.
[0084] In some embodiments, the side walls **21** of a part of the second spacer structures are all first side walls **21**A. In some embodiments, as illustrated in FIG. **6**, all the side walls **21** of the second spacer structure corresponding to or enclosing the second sub-pixel **10**B in the odd-numbered first pixel rows **110** are the first side walls **21**A.
[0085] As illustrated in FIG. **1**, FIG. **2** and FIG. **8**, FIG. **8** is a schematic structural diagram of a pixel structure according a seventh embodiment of the present disclosure.
[0086] A structure of the seventh embodiment of the present disclosure is substantially similar to that of the first embodiment. The difference between the seventh embodiment and the first embodiment is that: for a first spacer structure and a second spacer structure adjacent to each other along the second direction Y, the first side wall **21**A of this first spacer structure and the first side wall **21**A of this second spacer structure are disposed at opposite sides of the corresponding second symmetry line D**2**. The corresponding second symmetry line D**2** that corresponds to the first spacer structure is the second symmetry line D**2** of the column including the first sub-pixel **10**A enclosed by this first spacer structure, and the corresponding second symmetry line D**2** that corresponds to the second spacer structure is the second symmetry line D**2** of the column including the second sub-pixel **10**B enclosed by this second spacer structure. The first side walls **21**A adjacent to each other along the second direction Y are configured to contact each other and are electrically connected.
[0087] In some embodiments, for the first spacer structure and the second spacer structure adjacent to each other along the second direction Y, the first side wall **21**A of the first spacer structure and the first side wall **21**A of the second spacer structure are disposed on two opposite sides of the corresponding second symmetry line D**2**. The first side walls **21**A adjacent to each other along the second direction Y are configured to contact each other and are electrically connected.
[0088] In some embodiments, the second symmetry line D**2** includes a third side G**3** and a fourth side G**4** disposed opposite to each other along the first direction X. The third side G**3** of each second symmetry line D**2** is disposed at a same side of the second direction Y. For example, as illustrated in FIG. **8**, the third side G**3** is at the left side of the second direction Y.
[0089] For the first spacer structure and the second spacer structure adjacent to each other along the first direction X, the first side wall **21**A of the first spacer structure and the first side wall **21**A of the second spacer structure are disposed at the same side of the corresponding second symmetry lines D**2** respectively.
[0090] In some embodiments, in an odd-numbered first pixel row **110**, the first side wall **21**A of the first spacer structure is disposed on the third side G**3** of the second symmetry line D**2** corresponding to the first spacer structure, and the first side wall **21**A of the second spacer structure adjacent to the first spacer structure is also disposed on the third side G**3** of the second symmetry line D**2** corresponding to the second spacer structure. In an even-numbered first pixel row **110**, the first side wall **21**A of the first spacer structure is disposed on the fourth side G**4** of the second symmetry line D**2** corresponding to the first spacer structure, and the first side wall **21**A of the second spacer structure adjacent to the first spacer structure is also disposed on the fourth side G**4** of the second symmetry line D**2** corresponding to the second spacer structure.
[0091] As illustrated in FIG. **1**, FIG. **2** and FIG. **9**, FIG. **9** is a schematic structural diagram of a pixel structure according an eighth embodiment of the present disclosure.
[0092] In some embodiments, as illustrated in FIG. **9**, in an odd-numbered first pixel row **110**, the first side wall **21**A of the first spacer structure is disposed on the fourth side G**4** of the second

symmetry line D**2** corresponding to the first spacer structure, and the first side wall **21**A of the second spacer structure adjacent to the first spacer structure is also disposed on the fourth side G**4** of the second symmetry line D**2** corresponding to the second spacer structure. In an even-numbered first pixel row **110**, the first side wall **21**A of the first spacer structure is disposed on the third side G**3** of the second symmetry line D**2** corresponding to the first spacer structure, and the first side wall **21**A of the second spacer structure adjacent to the first spacer structure is also disposed on the third side G**3** of the second symmetry line D**2** corresponding to the second spacer structure.

[0093] As illustrated in FIG. **1**, FIG. **2**, FIG. **10** to FIG. **11**, FIG. **10** is a schematic structural diagram of a pixel structure according a ninth embodiment of the present disclosure, and FIG. **11** is a schematic structural diagram of a pixel structure according a tenth embodiment of the present disclosure.

[0094] In some embodiments, for the first spacer structure and the second spacer structure adjacent to each other along the first direction X, the first side wall **21**A of the first spacer structure and the first side wall **21**A of the second spacer structure are disposed on different sides of corresponding second symmetry lines D**2** respectively. In some embodiments, as illustrated in FIG. **10**, the first side wall **21**A of the first spacer structure is disposed on the third side G**3** of the second symmetry line D**2** corresponding to the first spacer structure, and the first side wall **21**A of the second spacer structure is disposed on the fourth side G**4** of the second symmetry line D**2** corresponding to the second spacer structure. In some embodiments, as illustrated in FIG. **11**, the first side wall **21**A of the first spacer structure is disposed on the fourth side G**4** of the second symmetry line D**2** corresponding to the first spacer structure, and the first side wall **21**A of the second spacer structure is disposed on the third side G**3** of the second symmetry line D**2** corresponding to the second spacer structure.

[0095] As illustrated in FIG. **1**, FIG. **2**, FIG. **12** to FIG. **14**, FIG. **12** is a schematic structural diagram of a pixel structure according an eleventh embodiment of the present disclosure, FIG. **13** is a schematic structural diagram of a pixel structure according a twelfth embodiment of the present disclosure, and FIG. **14** is a schematic structural diagram of a pixel structure according a thirteenth embodiment of the present disclosure.

[0096] In some embodiments, the side walls **21** of a part of the first spacer structures are all first side walls **21**A (not illustrated), and/or, the side walls **21** of a part of the second spacer structures are all first side walls **21**A, as illustrated in FIG. **12** and FIG. **13**. No more will be elaborated here. For more details, please refer to the above-mentioned descriptions.

[0097] In some embodiments, as illustrated in FIG. **14**, all the side walls **21** of the spacer structure **20** are the first side walls. In comparison with other embodiments of the present disclosure, a better electrical connection between the spacer structure **20** and the sub-pixel **10** may be achieved in this embodiment.

[0098] As illustrated in FIG. **1** to FIG. **15**, FIG. **15** is a schematic comparison diagram of a sub-pixel life decay process according to an embodiment of the present disclosure.

[0099] The above embodiments differ only in the specific arrangement location of the first side walls **21**A of the spacer structure **20**. In all these embodiments, the area of the pixel aperture **31** may be effectively increased, and the pixel current density may be reduced, thereby effectively enhancing the product lifespan, while further reducing the production cost of the pixel structures. In some embodiments, in comparison with the pixel structure of the related art prepared by an FMM process, the spacing between the sub-pixels **10** of the pixel structure of the present disclosure may be reduced by approximately 10 micrometers. In this way, provided a same pixel layout, the pixel aperture ratio of the pixel structure of the present disclosure may be approximately doubled as compared to the related art. In addition, as illustrated in FIG. **15**, the embodiment illustrated in the figure represents an embodiment of the present disclosure, and the comparison embodiment in the figure represents an embodiment of the related art without the spacer structure **20** of the present disclosure. As is illustrated in FIG. **15**, in the present disclosure, the lifespan capacity of the pixel structure is significantly improved in the present application as compared to the related art.

[0100] A display panel is provided in the present disclosure. The display panel includes any of the above-described pixel structures.

[0101] In the above-mentioned embodiments, a description of each embodiment has its own focus, and a part not detailed in a certain embodiment may be referred to relevant descriptions of other embodiments.

[0102] The above are only implementations of the present disclosure, and do not limit the patent scope of the present disclosure. Any equivalent changes to the structure or processes made by the description and drawings of this application or directly or indirectly used in other related technical field are included in the protection scope of this application.

Claims

1. A pixel structure, comprising a first sub-pixel, a second sub-pixel and a third sub-pixel, the colors of the first sub-pixel, the second sub-pixel and the third sub-pixel being different from each other, wherein the first sub-pixel and the second sub-pixel are alternately disposed along a first direction and form a first pixel row, a plurality of the third sub-pixels are disposed along the first direction and form a second pixel row; the first pixel row and the second pixel row are alternately disposed along a second direction, the first direction and the second direction are perpendicular to each other; two first sub-pixels and two second sub-pixels from adjacent two rows and adjacent two columns are arranged in a 2×2 matrix, the two first sub-pixels are located in different rows and different columns, the two second sub-pixels are located in different rows and different columns, two centers of the two first sub-pixels and the two centers of the two second sub-pixels are capable of being connected to form a virtual quadrilateral; the third sub-pixel is located within the virtual quadrilateral; a side of the third sub-pixel is parallel to an proximate side of an adjacent sub-pixel; the pixel structure further comprises spacer structures, the spacer structures are configured to enclose the sub-pixels, and are in one-to-one correspondence with the sub-pixels; the spacer structures are configured to electrically connect the cathodes of the sub-pixels to each other.

2. The pixel structure as claimed in claim 1, wherein each of the first sub-pixel and the second sub-pixel is a rhombus, the third sub-pixel is a parallelogram; a diagonal line of the rhombus is configured to extend in the first direction, another diagonal line of the rhombus is configured to extend in the second direction; the first sub-pixel and the second sub-pixel adjacent to each other along the first direction are configured to be disposed as corner-facing-corner, and the first sub-pixel and the second sub-pixel adjacent to each other along the second direction are configured to be disposed as corner-facing-corner; a side of the spacer structure is parallel to a proximate side of a sub-pixel adjacent to the spacer structure; the first sub-pixel and the third sub-pixel adjacent to the first sub-pixel share a side wall of the spacer structure, and the second sub-pixel and the third sub-pixel adjacent to the second sub-pixel share a side wall of the spacer structure.

3. The pixel structure as claimed in claim 2, wherein a region enclosed by the spacer structure is polygonal; corners of the region enclosed by the spacer structure configured to enclose the first sub-pixel are flat chamfers or rounded chamfers, and corners of the region enclosed by the spacer structure configured to enclose the second sub-pixel are flat chamfers or rounded chamfers.

4. The pixel structure as claimed in claim 2, wherein the spacer structure comprises a spacer part and an eave structure, the eave structure is provided on an upper surface of the spacer part; the eave structure is configured to cover the spacer part, and extend beyond the spacer part in a direction parallel to the spacer part; a cross-section of the spacer structure along a plane parallel to the eave structure is configured to taper along a direction toward the eave structure.

5. The pixel structure as claimed in claim 4, wherein for each spacer structure, at least a part of the side wall is a first side wall; the spacer part of the first side wall is electrically conductive, and is configured to contact the cathode of the sub-pixel adjacent to the first side wall, and achieve an electrical connection between the spacer structure and the sub-pixel; the cathode of the sub-pixel is vaporized onto a surface of the light-emitting layer of the sub-pixel by a vaporization source, the first direction is a long-edge direction of the vaporization source, and the second direction is a scanning direction of the vaporization source.

6. The pixel structure as claimed in claim 5, wherein a part of the side walls of each spacer structure is the first side wall, and another part of the side walls of the each spacer structure is a second side wall; and the spacer part of the second side wall is insulating.

7. The pixel structure as claimed in claim 5, wherein the spacer structure configured to enclose the first sub-pixel is referred to as a first spacer structure, and the spacer structure configured to enclose the second sub-pixel is referred to as a second spacer structure; a straight line on which the diagonal line of the rhombus extending in the first direction configured to extend is referred to as a first symmetry line, and a straight line on which the diagonal line of the rhombus extending in the second direction configured to extend is referred to as a second symmetry line; for the first spacer structure and the second spacer structure adjacent to each other along the first direction, the first side wall of the first spacer structure and the first side wall of the second spacer structure are disposed at opposite sides of the corresponding first symmetry line; the first side walls adjacent to each other along the first direction are configured to contact each other and are electrically connected; for the first spacer structure and the second spacer structure adjacent to each other along the second direction: the first side wall of the first spacer structure and the first side wall of the second spacer structure are disposed at the same side of the respectively corresponding first symmetry lines; or the first side wall of the first spacer structure and the first side wall of the second spacer structure are disposed at opposite sides of the respectively corresponding first symmetry lines.

8. The pixel structure as claimed in claim 5, wherein the spacer structure configured to enclose the first sub-pixel is referred to as a first spacer structure, and the spacer structure configured to enclose the second sub-pixel is referred to as a second spacer structure; a straight line on which the diagonal line of the rhombus extending in the first direction configured to extend is referred to as a first symmetry line, and a straight line on which the diagonal line of the rhombus extending in the second direction configured to extend is referred to as a second symmetry line; for the first spacer structure and the second spacer structure adjacent to each other along the second direction, the first side wall of the first spacer structure and the first side wall of the second spacer structure are disposed at opposite sides of the corresponding second symmetry line; the first side walls adjacent to each other along the second direction are configured to contact each other and are electrically connected; for the first spacer structure and the second spacer structure adjacent to each other along the first direction: the first side wall of the first spacer structure and the first side wall of the second spacer structure are disposed at the same side of the respectively corresponding second symmetry lines; or the first side wall of the first spacer structure and the first side wall of the second spacer structure are disposed at opposite sides of the respectively corresponding second symmetry lines.

9. The pixel structure as claimed in claim 7, wherein the side walls of a part of the first spacer structures are all first side walls; and/or the side walls of a part of the second spacer structures are all first side walls.

10. The pixel structure as claimed in claim 8, wherein the side walls of a part of the first spacer structures are all first side walls; and/or the side walls of a part of the second spacer structures are all first side walls.

11. A display panel comprising a pixel structure, wherein the pixel structure comprises a first sub-pixel, a second sub-pixel and a third sub-pixel, and the colors of the first sub-pixel, the second sub-pixel and the third sub-pixel are different from each other; the first sub-pixel and the second sub-pixel are alternately disposed along a first direction and form a first pixel row, a plurality of the third sub-pixels are disposed along the first direction and form a second pixel row; the first pixel row and the second pixel row are alternately disposed along a second direction, the first direction and the second direction are perpendicular to each other; two first sub-pixels and two second sub-pixels from adjacent two rows and adjacent two columns are arranged in a 2×2 matrix, the two first sub-pixels are located in different rows and different columns, the two second sub-pixels are located in different rows and different columns, two centers of the two first sub-pixels and the two centers of the two second sub-pixels are capable of being connected to form a virtual quadrilateral; the third sub-pixel is located within the virtual quadrilateral; a side of the third sub-pixel is parallel to a proximate side of an adjacent sub-pixels; the pixel structure further comprises spacer structures, the spacer structures are configured to enclose the sub-pixels, and are in one-to-one correspondence with the sub-pixels; the spacer structures are configured to electrically connect the cathodes of the sub-pixels to each other.

12. The display panel as claimed in claim 11, wherein each of the first sub-pixel and the second sub-pixel is a rhombus, the third sub-pixel is a parallelogram; a diagonal line of the rhombus is configured to extend in the first direction, another diagonal line of the rhombus is configured to extend in the second direction; the first sub-pixel and the second sub-pixel adjacent to each other along the first direction are configured to be disposed as corner-facing-corner, and the first sub-pixel and the second sub-pixel adjacent to each other along the second direction are configured to be disposed as corner-facing-corner; a side of the spacer structure is parallel to a proximate side of a sub-pixel adjacent to the spacer structure; the first sub-pixel and the third sub-pixel adjacent to the first sub-pixel share a side wall of the spacer structure, and the second sub-pixel and the third sub-pixel adjacent to the second sub-pixel share a side wall of the spacer structure.

13. The display panel as claimed in claim 12, wherein a region enclosed by the spacer structure is polygonal; corners of the region enclosed by the spacer structure configured to enclose the first sub-pixel are flat chamfers or rounded chamfers, and corners of the region enclosed by the spacer structure configured to enclose the second sub-pixel are flat chamfers or rounded chamfers.

14. The display panel as claimed in claim 12, wherein the spacer structure comprises a spacer part and an eave structure, the eave structure is provided on an upper surface of the spacer part; the eave structure is configured to cover the spacer part, and extend beyond the spacer part in a direction parallel to the spacer part; a cross-section of the spacer structure along a plane parallel to the eave structure is configured to taper along a direction toward the eave structure.

15. The display panel as claimed in claim 14, wherein for each spacer structure, at least a part of the side wall is a first side wall; the spacer part of the first side wall is electrically conductive, and is configured to contact the cathode of the sub-pixel adjacent to the first side wall, and achieve an electrical connection between the spacer structure and the sub-pixel; the cathode of the sub-pixel is vaporized onto a surface of the light-emitting layer of the sub-pixel by a vaporization source, the first direction is a long-edge direction of the vaporization source, and the second direction is a scanning direction of the vaporization source.

16. The display panel as claimed in claim 15, wherein a part of the side walls of each spacer structure is the first side wall, and another part of the side walls of the each spacer structure is a second side wall; and the spacer part of the second side wall is insulating.

17. The display panel as claimed in claim 15, wherein the spacer structure configured to enclose the first sub-pixel is referred to as a first spacer structure, and the spacer structure configured to enclose the second sub-pixel is referred to as a second spacer structure; a straight line on which the diagonal line of the rhombus extending in the first direction configured to extend is referred to as a first symmetry line, and a straight line on which the diagonal line of the rhombus extending in the second direction configured to extend is referred to as a second symmetry line; for the first spacer structure and the second spacer structure adjacent to each other along the first direction, the first side wall of the first spacer structure and the first side wall of the second spacer structure are disposed at opposite sides of the corresponding first symmetry line; the first side walls adjacent to each other along the first direction are configured to contact each other and are electrically connected; for the first spacer structure and the second spacer structure adjacent to each other along the second direction: the first side wall of the first spacer structure and the first side wall of the second spacer structure are disposed at the same side of the respectively corresponding first symmetry lines; or the first side wall of the first spacer structure and the first side wall of the second spacer structure are disposed at opposite sides of the respectively corresponding first symmetry lines.

18. The display panel as claimed in claim 15, wherein the spacer structure configured to enclose the first sub-pixel is referred to as a first spacer structure, and the spacer structure configured to enclose the second sub-pixel is referred to as a second spacer structure; a straight line on which the diagonal line of the rhombus extending in the first direction configured to extend is referred to as a first symmetry line, and a straight line on which the diagonal line of the rhombus extending in the second direction configured to extend is referred to as a second symmetry line; for the first spacer structure and the second spacer structure adjacent to each other along the second direction, the first side wall of the first spacer structure and the first side wall of the second spacer structure are disposed at opposite sides of the corresponding second symmetry line; the first side walls adjacent to each other along the second direction are configured to contact each other and are electrically connected; for the first spacer structure and the second spacer structure adjacent to each other along the first direction: the first side wall of the first spacer structure and the first side wall of the second spacer structure are disposed at the same side of the respectively corresponding second symmetry lines; or the first side wall of the first spacer structure and the first side wall of the second spacer structure are disposed at opposite sides of the respectively corresponding second symmetry lines.

19. The display panel as claimed in claim 17, wherein the side walls of a part of the first spacer structures are all first side walls; and/or the side walls of a part of the second spacer structures are all first side walls.

20. The display panel as claimed in claim 18, wherein the side walls of a part of the first spacer structures are all first side walls; and/or the side walls of a part of the second spacer structures are all first side walls.